

TechBiz Security

Cyber Security Internship Program

Week 4 – Task 10 (SOC)

SIEM Fundamentals and Wazuh SIEM Implementation

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1. Introduction

Security Information and Event Management (SIEM) refers to a class of security platforms that collect, centralize, and analyse security-related data generated across an organisation's IT environment. A SIEM platform gathers logs and events from multiple sources such as servers, endpoints, applications, and network devices, and brings them into a single system where they can be correlated, searched, and reviewed by security analysts.

Within a Security Operations Centre (SOC), SIEM platforms play a central role because they allow analysts to move away from checking individual systems one at a time and instead work from a unified view of security activity across the environment. This centralised visibility supports security monitoring, event analysis, threat detection, investigation of suspicious behaviour, and the overall incident response process.

This report documents the practical work carried out for Week 4 – Task 10, which focused on understanding SIEM fundamentals and gaining hands-on exposure to Wazuh, an open-source SIEM and security monitoring platform. The task also required completion of the TryHackMe “Introduction to SIEM” learning room to reinforce the theoretical concepts alongside the practical dashboard review.

2. Objectives

1. Understand the fundamental concept of SIEM.
2. Understand security log collection and ingestion.
3. Understand event correlation and security alerting.
4. Understand the role of SIEM dashboards in SOC monitoring.
5. Gain practical exposure to Wazuh SIEM.
6. Verify the Wazuh SIEM dashboard and services.
7. Complete the TryHackMe Introduction to SIEM learning room.
8. Document the practical work and evidence for internship evaluation.

3. SIEM Fundamentals

3.1 Log Collection

SIEM platforms collect logs and security events from a wide range of sources, including endpoint devices, servers, applications, network appliances, and dedicated security tools such as firewalls and intrusion detection systems. Each of these sources generates records of activity that, on their own, provide only a partial picture of what is happening in the environment. A SIEM platform pulls this information together so that it can be reviewed in one place.

3.2 Log Ingestion

Once logs are generated at the source, they need to be transferred into the SIEM platform for storage and analysis. This process, known as ingestion, typically involves lightweight agents or forwarders installed on monitored systems that send log data to the central SIEM server, where it is parsed, indexed, and made searchable.

3.3 Event Correlation

Event correlation is the process by which a SIEM platform examines multiple related events together, rather than in isolation, to identify patterns that may indicate suspicious or malicious activity. A single event viewed on its own may appear harmless, but when correlated with other events across time or across systems, it can reveal behaviour that warrants further investigation.

3.4 Alerting

SIEM platforms generate alerts when incoming events match predefined detection rules or exceed certain severity thresholds. Alerts are typically categorised by severity level, which helps analysts prioritise which events require immediate attention and which can be reviewed as part of routine monitoring.

3.5 Dashboards and Visualization

Dashboards give security analysts a centralised, visual overview of the environment, summarising information such as active alerts, connected agents, and overall security posture. Rather than searching through raw logs manually, analysts can use dashboards to quickly identify areas that need closer review.

4. Key Benefits of SIEM

4.1 Centralized Log Management

By bringing logs from multiple sources into a single platform, SIEM removes the need to check each system individually, giving SOC teams centralised visibility over security-related activity across the organisation.

4.2 Threat Detection and Alerting

SIEM platforms apply detection rules to incoming data and generate alerts when suspicious activity is identified, allowing analysts to focus their attention on events that carry genuine security significance.

4.3 Security Monitoring and Investigation

Dashboards and searchable event data allow SOC analysts to monitor the environment continuously and to investigate specific incidents by tracing related events and reviewing supporting evidence.

In addition to these core benefits, SIEM platforms also support broader activities such as incident response, compliance-related monitoring, and general security visibility across the organisation, though these were not directly exercised as part of this task.

5. Technology Used

Technology / Platform	Purpose
Wazuh	Open-source SIEM and security monitoring platform used for practical implementation
TryHackMe	Used to study and complete the Introduction to SIEM learning room
Ubuntu	Operating system hosting the Wazuh server environment
VirtualBox	Virtualization platform used to run the Wazuh server virtual machine
Web Browser	Used to access the Wazuh Dashboard

6. Wazuh SIEM Environment

For this task, an existing Wazuh environment, already deployed on a Ubuntu virtual machine within VirtualBox, was used rather than performing a fresh installation. The purpose of this task was to verify the running environment, access the dashboard, and understand the SIEM workflow at a practical level.

At a high level, a Wazuh deployment consists of the following core components:

- Wazuh Manager – the central component responsible for analysing data received from agents and applying detection rules.
- Wazuh Indexer – responsible for storing and indexing the security data so that it can be searched and queried efficiently.
- Wazuh Dashboard – the web-based interface that presents security information, alerts, and monitoring modules to the analyst.

These components work together to provide the end-to-end SIEM workflow, from data collection through to visualisation. No additional configuration changes were made to these components as part of this task; the work was limited to verifying that the services were running and reviewing the dashboard.

7. Wazuh Service Verification

The existing Wazuh services were started and verified on the server using standard systemd service management commands. The following commands were used:

```
sudo systemctl start wazuh-manager
sudo systemctl start wazuh-indexer
sudo systemctl start wazuh-dashboard
```

Following this, the status of the Wazuh Dashboard service was checked to confirm that it was active and running before proceeding to access the web interface.

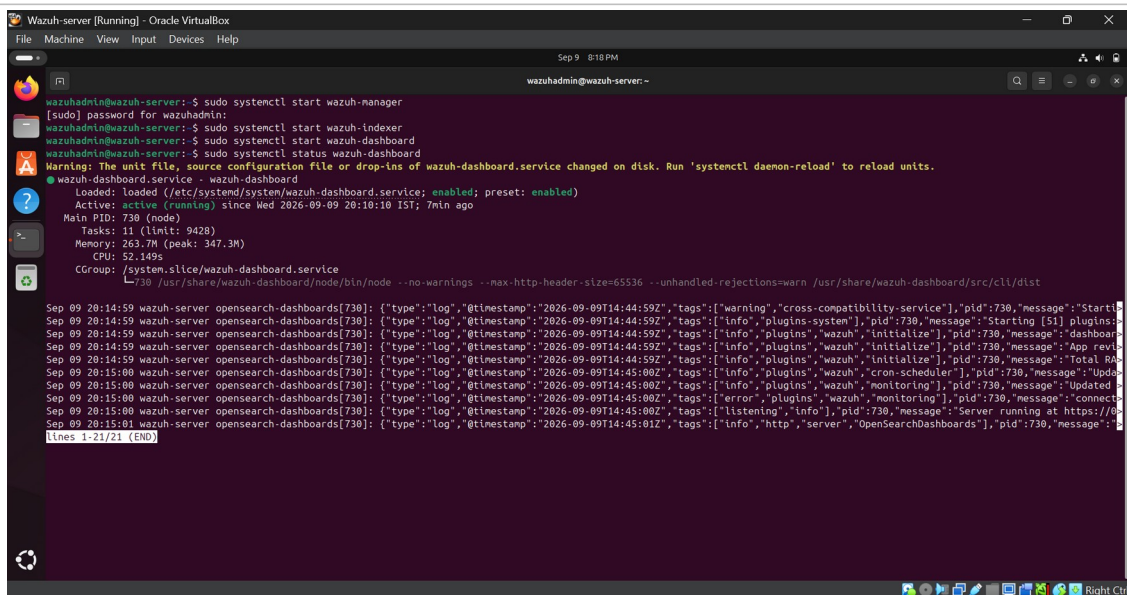


Figure 1: Wazuh Dashboard Service Running

The screenshot above provides evidence that the Wazuh Dashboard service was running successfully on the Wazuh server, confirming that the platform was in a suitable state to be accessed through a web browser.

8. Wazuh Server Network Identification

Before the Wazuh web dashboard could be accessed, the IP address assigned to the Wazuh server was identified. This was obtained by running the following command on the server:

```
hostname -I
```

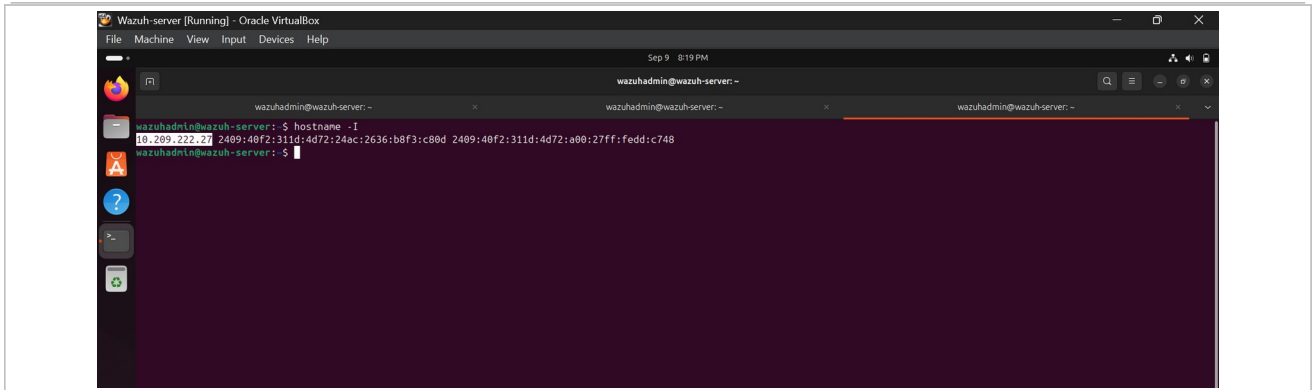


Figure 2: Wazuh Server IP Address

The screenshot above shows the IP address assigned to the Wazuh server, which was subsequently used in a web browser to access the Wazuh dashboard.

9. Wazuh Dashboard Access

Using the identified server IP address, the Wazuh web dashboard was accessed through a web browser. The dashboard serves as a centralised interface for security monitoring and provides access to a number of functional areas, including agent management, alerting, endpoint security, threat intelligence, security operations, and cloud security modules.

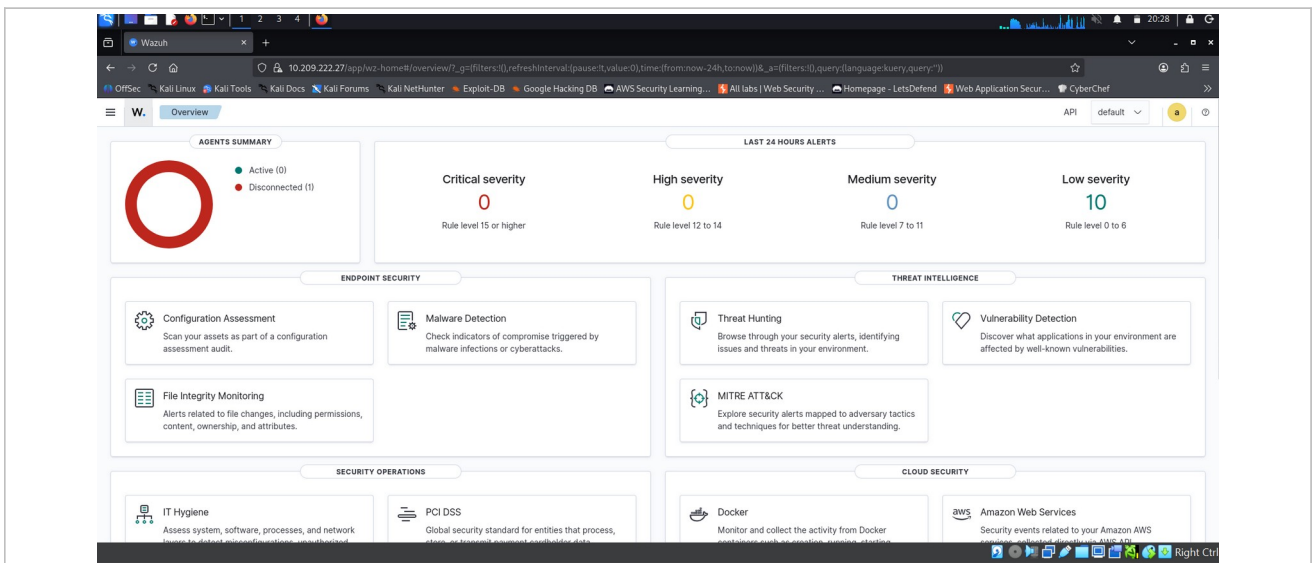


Figure 3: Wazuh SIEM Dashboard Overview

The dashboard overview screenshot shows an Agents Summary indicating 0 active agents and 1 disconnected agent. The Last 24 Hours Alerts panel shows 0 critical severity alerts, 0 high severity alerts, 0 medium severity alerts, and 10 low severity alerts. The dashboard also displays the available Endpoint Security modules (Configuration Assessment, Malware Detection, File Integrity Monitoring), Threat Intelligence modules (Threat Hunting, Vulnerability Detection, MITRE ATT&CK), Security Operations modules (IT Hygiene, PCI DSS), and Cloud Security modules (Docker, Amazon Web Services). No endpoint agent was actively connected at the time this screenshot was captured, and the low severity alerts shown do not correspond to any specific identified attack scenario.

10. Wazuh Login Environment

Prior to interacting with the Wazuh dashboard, the Wazuh server environment was accessed through its Ubuntu-based login screen, confirming that the underlying virtual machine hosting the Wazuh services was running and available.

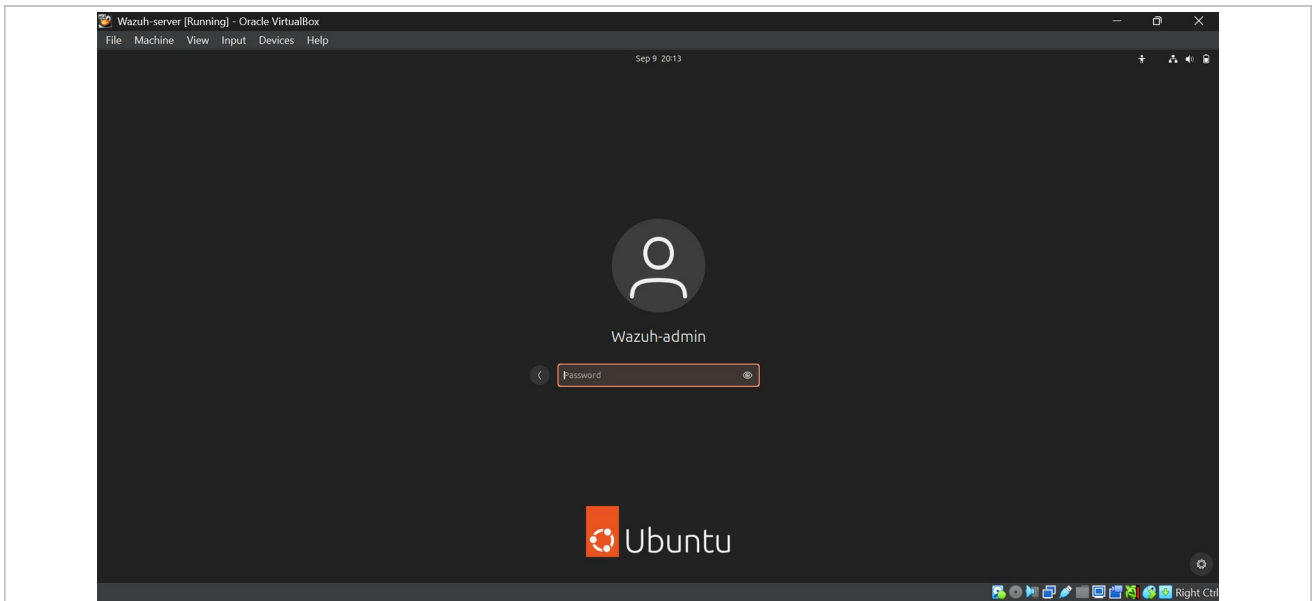


Figure 4: Wazuh Server Login Screen

11. TryHackMe – Introduction to SIEM

As part of the theoretical learning requirement for this task, the TryHackMe “Introduction to SIEM” room was completed. This learning room covers a set of foundational SIEM concepts, including:

- SIEM fundamentals
- Log collection
- Log sources and ingestion
- Alerting and analysis
- The purpose and functionality of SIEM platforms

These concepts directly supported the practical review of the Wazuh dashboard carried out in this task, providing the background needed to interpret the information presented in the dashboard modules.

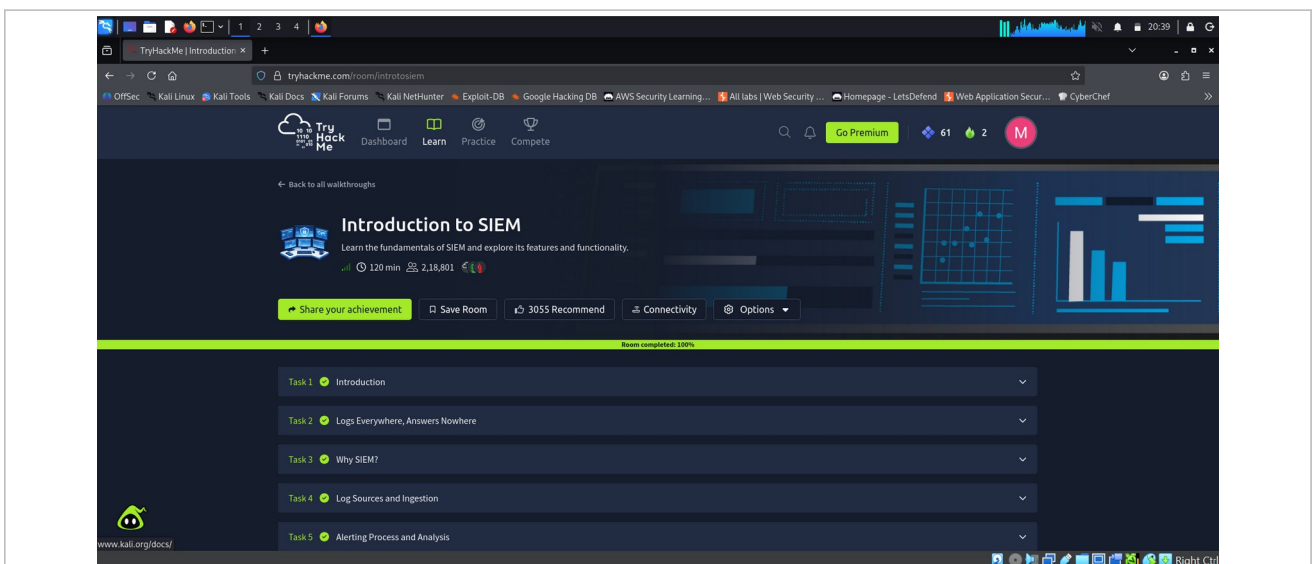


Figure 5: TryHackMe Introduction to SIEM – 100% Completion

The screenshot above shows the TryHackMe room completion status at 100 percent, confirming that the required SIEM learning module was completed in full.

12. Practical Observations

- The Wazuh Dashboard service was running successfully.
- The Wazuh server IP address was identified.
- The Wazuh Dashboard was accessible through a web browser.
- The Wazuh Dashboard provided centralised visibility into security monitoring modules.
- The dashboard displayed alert severity information, including 10 low severity alerts in the last 24 hours.
- The TryHackMe Introduction to SIEM room was completed at 100 percent.
- The practical environment demonstrated the basic workflow of accessing and reviewing a SIEM dashboard.

It is noted that the Agents Summary displayed 0 active agents and 1 disconnected agent, meaning no endpoint agent was actively reporting data to the Wazuh Manager during this session. As such, this task should be understood as a review of the SIEM dashboard and its available modules rather than a demonstration of active endpoint monitoring.

13. Security Analyst / SOC Relevance

The knowledge and skills exercised in this task are directly relevant to the day-to-day responsibilities of a SOC analyst. In practice, a SOC analyst uses a SIEM platform such as Wazuh to:

- Monitor security events across the environment
- Review alerts generated by detection rules
- Identify suspicious activity requiring further review
- Investigate security events using available log data
- Correlate information across multiple sources
- Prioritise alerts based on severity level
- Support the broader incident response process

For an entry-level Cyber Security Intern, developing familiarity with how to navigate a SIEM dashboard, interpret agent status, and read alert severity summaries forms a practical foundation for taking on SOC monitoring responsibilities.

14. Challenges and Learning

This task provided practical understanding of several core SOC concepts, including SIEM architecture, log collection concepts, event analysis, alert severity classification, security dashboards, the Wazuh interface, and the general SOC monitoring workflow. Working through the TryHackMe learning room alongside the Wazuh dashboard review helped connect the theoretical concepts to a real platform interface.

15. Conclusion

This task successfully provided foundational knowledge of SIEM concepts and practical exposure to the Wazuh platform. Completing the TryHackMe “Introduction to SIEM” room alongside the verification of the Wazuh

services and review of the Wazuh dashboard helped demonstrate how SIEM technology is used to provide centralised security monitoring and analysis within a SOC environment.

16. Evidence Summary

Evidence No.	Evidence	Purpose
1	Wazuh Server Login Screen	Shows the Wazuh server environment
2	Wazuh Dashboard Service Running	Verifies the Wazuh Dashboard service
3	Wazuh Server IP Address	Shows the server address used for dashboard access
4	Wazuh Dashboard Overview	Demonstrates access to the SIEM dashboard and visible security information
5	TryHackMe Introduction to SIEM Completion	Demonstrates 100% completion of the SIEM learning room

17. Final Learning Outcomes

1. Understood the fundamentals of SIEM.
2. Learned the importance of centralized security logging.
3. Understood log ingestion and event correlation.
4. Learned the purpose of SIEM alerting and dashboards.
5. Gained practical exposure to Wazuh.
6. Verified access to the Wazuh SIEM dashboard.
7. Completed the TryHackMe Introduction to SIEM room.
8. Improved understanding of the basic SOC monitoring workflow.